

Hugo Pereira

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EDUCATION

Instituto Superior Técnico, University of Lisbon Master of Science in Aerospace Engineering (Control & Systems branch)	2020 - 2022 GPA: 19 / 20
Technical University of Denmark Exchange Student	2020 - 2021 GPA: 11 / 12
Instituto Superior Técnico, University of Lisbon Bachelor of Science in Aerospace Engineering	2017 - 2020 GPA: 17 / 20

RESEARCH EXPERIENCE

 Institute for Systems and Robotics, IST-UL <i>Graduate Student Researcher (Full-time)</i>	<i>Lisbon, Portugal</i> <i>Mar. 2022 - June 2023</i>
• Research on singularity avoidance for spacecraft attitude control with control moment gyros (CMGs). • Derived a novel convex MPC-based solution for CMG allocation. • Developed a medium-fidelity 6 DoF simulator for spacecraft dynamics in LEO. • Worked towards my master's thesis titled "<i>A Convex Allocation Framework for Singularity Avoidance in Control Moment Gyro Clusters</i>", which was awarded the grade of 20 out of 20.	

WORK EXPERIENCE

 GMV <i>Control systems engineer (Full-time)</i>	<i>Lisbon, Portugal</i> <i>July 2023 - Present</i>
• Responsible for the design and development of the attitude controllers for the Rapid Apophis Mission for Space Safety (RAMSES) through PDR and CDR phases. Carried out control synthesis and frequency-domain analysis for all the control modes of the GNC (S/C detumbling, Sun pointing operations during Safe mode, RW and RCT control for attitude stabilization and slew maneuvers, and wheel off-loading). • Co-developed the attitude controllers for the ARIEL telescope. Worked out a solution for attitude estimation by combining star tracker data with fine guidance sensor measurements. • Worked as a system engineer during the development of a testbed for single-axis attitude control of a spacecraft with challenging flexible dynamics. Co-developed an in-house reaction wheel, designed the avionics for the platform, handled all the hardware procurement, and took responsibility in the development of a digital twin using MATLAB/SIMULINK. • Participated in several stages of research-oriented ESA-funded activities on spacecraft collision avoidance (literature review and requirements definition) and space launchers (derivation of flight dynamics models and synthesis of convex methods for thruster allocation). • Co-supervised and oversaw the work of five interns on the topics of convex optimization, control theory, and system identification applied to flexible space structures.	

- Served as an invited lecturer for the Convex Optimization course, delivering 6 hrs/week of problem-solving sessions to MSc classes. Helped students with the project work and assisted in grading exams.

- Short-term internship where I contributed to the design of a signal demodulation program for an Emergency Position Indicating Radio Beacon. The system, developed in collaboration with other interns, was sketched in GNU Radio and written in Python.

EXTRACURRICULAR ACTIVITIES

- Developed autonomous driving algorithms, using ROS/C++, for a driverless Formula car. The software pipeline encompasses machine learning for point cloud processing, state estimation via Kalman filtering, simultaneous localization and mapping (SLAM), and model predictive control (MPC).
- In specific, I implemented new data association methods for SLAM, developed an iterative closest point (ICP) method for point cloud alignment, implemented point cloud processing algorithms, and designed and fine-tuned longitudinal and lateral nonlinear controllers for path tracking.
- Achieved a 3rd place at Formula Student Germany 2022 and a 1st at Formula Student Spain 2022 (driverless categories).

- Part of the student team that set the altitude world record for collegiate bi-liquid rocketry. Apogee: 6545 m AGL.
- Led the development of the rocket's telemetry system. Established a communication link based on LoRa technology that operated at a 9 km range. Designed and tested PCBs for radio communication, using KiCad, within an STM32-based flight computer architecture.
- Carried out robust flight simulations using the OpenRocket software, which resulted in a predicted apogee error of less than 3%.
- Achieved a 1st place at the European Rocketry Challenge 2021 (bi-liquid propulsion category).

- Built a UAV from scratch over the course of two years. Initially, I was involved in the design, 3D modelling, and construction of the UAV. At a later stage, I joined the development of the flight controller, which involved PID control, filtering, path planning, sensor fusion, and parameter identification. The simulations were conducted in MATLAB/SIMULINK and the on-board code was implemented in an RPi.
- Gave workshops about rocketry to high school students. These consisted of a hands-on experience that included planning, building, and launching a small rocket made from cheap conventional materials.

PUBLICATIONS

H. Pereira, P. Lourenço, P. Batista, *From singularity analysis to singularity avoidance: novel metric and convex allocation for spacecraft attitude control with control moment gyros*, Acta Astronautica 225 (2024) 41-54, doi: 10.1016/j.actaastro.2024.08.047.

[MSc thesis] H. Pereira, *A Convex Allocation Framework for Singularity Avoidance in Control Moment Gyro Clusters*, Instituto Superior Técnico, 2022.

SKILLS SUMMARY

Languages	English (C2), Portuguese (Native), Spanish (B1)
Programming	C++, C, MATLAB, Python
Software	SIMULINK, KiCad, OpenRocket, SolidWorks
Tools	Git, LATEX, Microsoft Office, Jira
Platforms	Linux, Windows, ROS, STM32, PIC24, Arduino
Miscellaneous	Soldering, Driving License

COMPLEMENTARY TRAINING

- *Inter-Agency GNC V&V Workshop (IAGNC 2025), ISAE-SUPAERO, July. 2025*. Short training course on H_∞ control and dynamic systems modeling and verification applied to spacecraft attitude control.
- *Ladybird Guide to Spacecraft Communications Training Course, 2022, ESA Academy*. Selected to attend a remote spacecraft communications course featuring live sessions given by ESA experts.
- *Summer course in Robotics, University of Twente, Aug. 2019*. On-site summer camp in Robotics at the University of Twente in the Netherlands. This consisted of a hands-on experience with real-time robotics systems and included a group project centered on Arduino manipulation.

VOLUNTEERING

Formula Student Portugal, FSPT25

July. 2025

- Acted as a design judge for the driverless system, evaluating and providing feedback on teams' autonomous navigation solutions, including SLAM implementation and Kalman filter design.

Erasmus Student Network, DTU section

Feb. 2021 - June 2021

- Member of a multinational team of volunteers whose goal is to facilitate the integration of international students in the local community.
- Organized social events and provided general support and guidance for exchange students.

HONORS AND AWARDS

- Academic Excellence, Instituto Superior Técnico, 2021/2022
- Outstanding Performance at EuRoc 2021, DTU Blue Dot, Oct. 2021
- Academic Merit, Instituto Superior Técnico, 2017/2018, 2019/2020
- Best Student Award, Escola Secundária Marquesa de Alorna, 2015/2016, 2016/2017